

- There are 3 Questions for 12 marks.
- Write your answers neatly, legibly and logically. Keep your proofs thorough. Your proofs must follow from the basics discussed in class.
- You do not need any calculators.
- Good luck!

1. [2] What is the limit of the following sequences

(a) $(1 + 1/2n)^n$

(b) $(-1)^n n! / n^n$

2. [4] What is the radius of convergence and limit of the following series

(a) $1 - x + x^2 - x^3 + x^4 - \dots$

(b) $\sum_{-\infty}^0 2^i x^i$

(c) $\sum_{-\infty}^0 a_i x^i$ where $a_i = \begin{cases} 2^i & \text{for } i < 0 \\ (-1)^i & \text{otherwise} \end{cases}$

3. (a) [2] Using the series expansion of $1/(1+x)$, derive the Taylor series for $\ln(1+x)$ about $x=0$.

(b) [2] What is the radius of convergence of this series ?

(c) [2] Give an approximation for $\ln(1.5)$ up to the second decimal place (you may use, if required:
 $1/64 \approx 0.015, 1/160 \approx 0.006$).

$$1) (a) \lim \left(1 + \frac{1}{2n}\right)^n = \lim \left[\left(1 + \frac{1}{2n}\right)^{2n} \right]^{\frac{1}{2}}$$

Since $\left(1 + \frac{1}{2n}\right)^{2n} \rightarrow e$, it follows that $\lim \left(1 + \frac{1}{2n}\right)^n = e^{1/2}$

$$(b) (-1)^n \frac{n!}{n^n}$$

$\frac{n!}{n^n} = \frac{1}{n} \cdot \frac{2}{n} \cdot \frac{3}{n} \cdots \frac{n-1}{n} \cdot \frac{n}{n} \rightarrow 0$ can be proved multiple ways

Approach 1: In class, it was proved that $\frac{n^n}{n!} \rightarrow \infty$. Similar techniques can be used here.

Take the ratio of successive terms: $\frac{(n+1)!}{(n+1)^{n+1}} \bigg/ \frac{n!}{n^n} = \frac{1}{\left(1 + \frac{1}{n}\right)^n} < 1$,
 $\approx \frac{1}{e}$ when n is large. ①

Thus $\frac{n!}{n^n}$ is a decreasing sequence bounded below (by 0) and hence has a limit ≥ 0 . If $\lim \frac{n!}{n^n} = L > 0$

then $\lim \frac{(n+1)!}{n^{n+1}} \bigg/ \frac{n!}{n^n}$ must be $\frac{L}{L} = 1$; However it is $1/e$ from the previous calculation (in ①). Thus we must have

$$\lim \frac{n!}{n^n} = 0$$

Approach 2: Use Stirling's approximation

$$\ln n! = n \ln n - n + O(\ln n)$$

$$\text{thus } \ln \frac{n!}{n^n} = n \ln n - n + O(\ln n) - n \ln n = -n + O(\ln n)$$

$$\text{i.e. } \frac{n!}{n^n} \leq e^{-n} n^c \text{ for constant } c$$

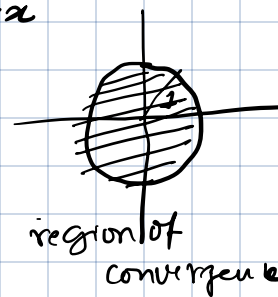
Taking limits, $\lim \frac{n!}{n^n} = 0$.

2

(a) $1 - x + x^2 - x^3 \dots$

this is a Geometric series, and as discussed in class, converges only when $|x| < 1$. the limit is $\frac{1}{1+x}$.

The radius of convergence is 1.

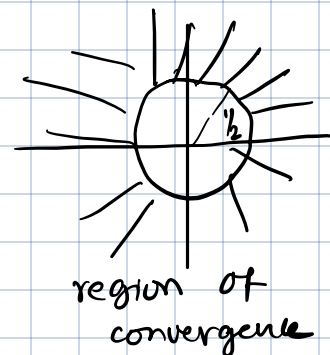


(b) $\sum_{i=-\infty}^0 2^i x^i = 1 + \frac{1}{2x} + \frac{1}{(2x)^2} + \dots$

Is a Geometric series as well, with limit $\frac{1}{1 - \frac{1}{2x}} = \frac{2x}{2x-1}$

and converges only when $|\frac{1}{2x}| < 1$ i.e $|x| > \frac{1}{2}$.

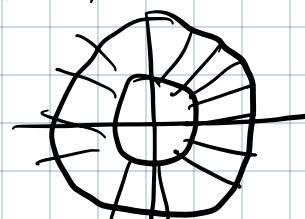
So the region of convergence is outside disc of radius $\frac{1}{2}$



(c) this is the sum (except one term) of the previous two series, with

limit $\frac{1}{1+x} + \frac{2x}{2x-1} - 1$ for $\frac{1}{2} < |x| < 1$. the region

of convergence is the intersection of the previous regions



$$\underline{3(a)} \quad \frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots$$

Integrating both sides,

$$\ln(1+x) = \text{const} + x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \dots$$

Set $x = 0$ to obtain $\text{const} = 0$.

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \quad \text{--- (2)}$$

(b) the region of convergence is the same as that of the original series i.e. $|x| < 1$. We can alternately do ratio test on (2) to obtain the region of convergence

$$\text{By ratio test} \quad \lim \left| \frac{a_{n+1}}{a_n} \right| = \lim \left| \frac{x^{n+1}/(n+1)}{x^n/n} \right| = |x|$$

So for $|x| < 1$, the series converges.

(c) Use $x = 0.5$,

$$\ln(1.5) = \frac{1}{2} - \frac{1}{8} + \frac{1}{24} - \frac{1}{64} + \frac{1}{160} - \dots$$

Since this is an alternating series, the error in estimating upto the k^{th} term is at most the $(k+1)^{\text{th}}$ term.

$$\begin{aligned} \ln(1.5) &\approx \frac{1}{2} - \frac{1}{8} + \frac{1}{24} - \frac{1}{64} \quad \text{with error at most } \frac{1}{160} \approx 0.006 \\ &\approx 0.40 \quad \text{with error at most } 0.006 \end{aligned}$$